1. Data Understanding

Dataset Structure and Overview

The dataset contains a sample of 5000 customers with 20 variables, including identities (*customer_id*), categorical variables (*region*, *membership_tier*, *acquisition_channel*), and numerical metrics for engagement (*tenure_month*, *recency_days*), spending (*orders_12m*, *avg_basket*), and behavioral traits (channel and category shares).

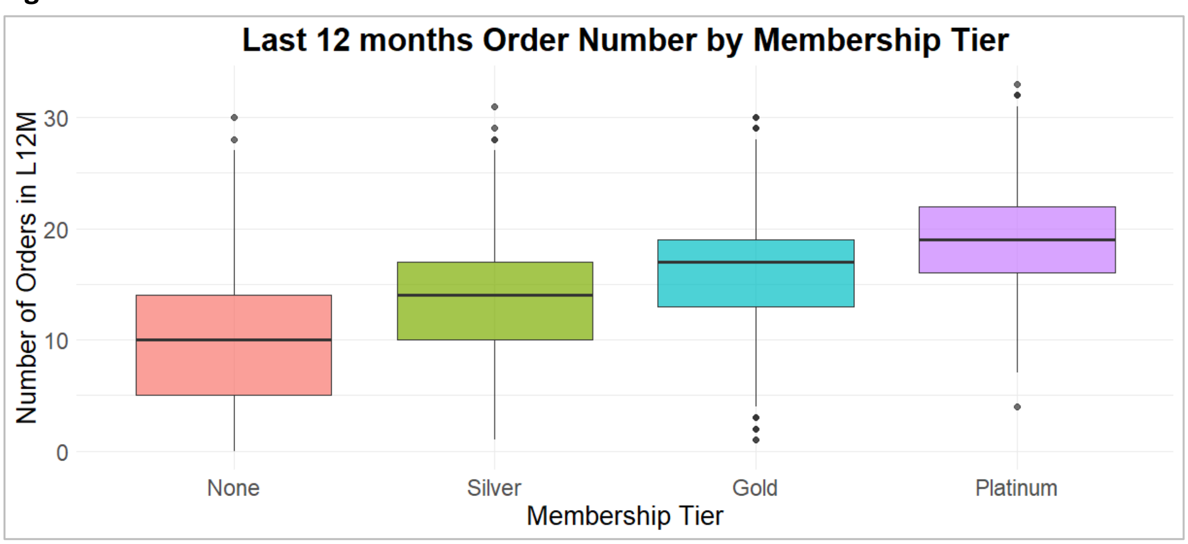
Potential Issues

1. **Missing Values (NAs):**
   The *delivery_delay_mean* variable contains 309 missing values.
2. **Scale Variance and Skewness:**
   The spend variable total_spend_12m is right-skewed, with a range of 7574, while the preference variables are bounded between 0 and 1, which could cause the spend variables to dominate and skew the clustering.
3. **Over-segmenting:**
   With 20 variables in the dataset, there is a risk of over-segmenting, where clusters become indistinguishable.

2. Exploratory Data Visualization

1. Order Frequency by Membership Tier

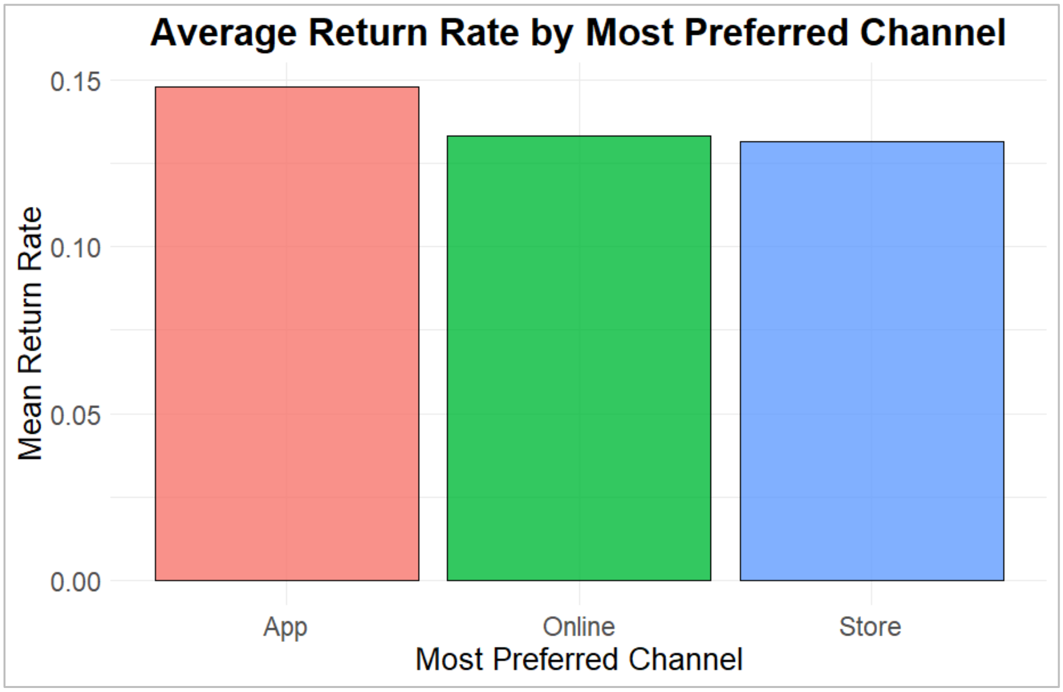
Figure 1



The boxplot compares purchase frequency across the four membership tiers.
Order frequency increases as membership status rises from “None” to “Platinum”, indicating that the loyalty program incentivizes frequent purchases. However, there are customers in the “None” tier who are high-frequency outliers, whose purchase behaviour is almost similar to the ones in “Silver” or “Gold.

2. Average Return Rate by Most Preferred Channel

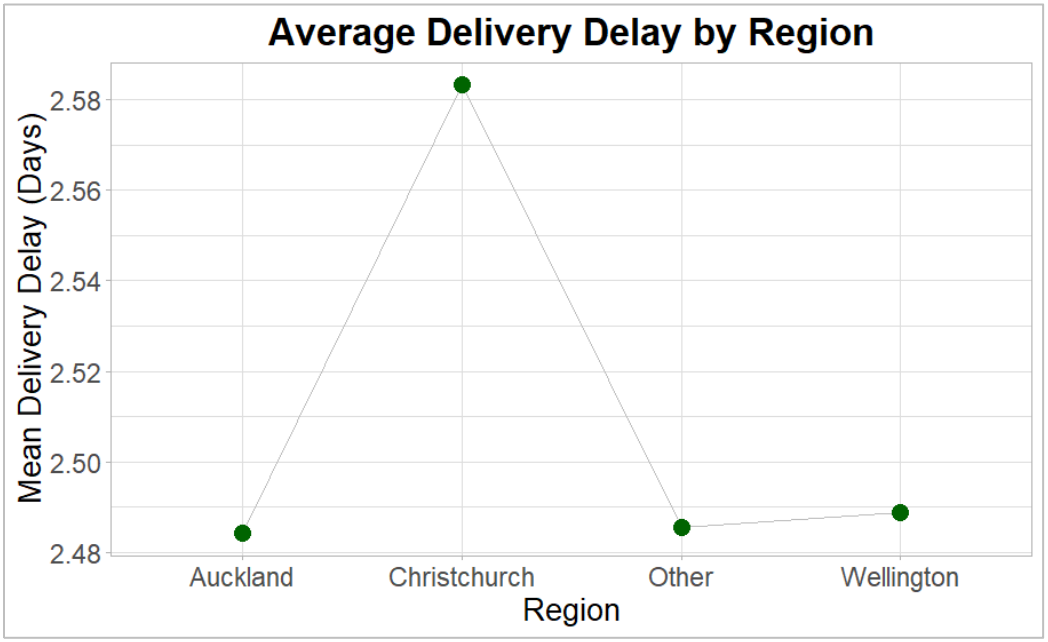
Figure 2



The bar chart displays the mean return rate for each of the most preferred channels (the channel with the highest rate for each customer).
The app channel has the highest return rate, averaging 15%, while “Online” is the second-highest. This suggests virtual customers are more likely to return orders.

3. Average Delivery Delay by Region

Figure 3

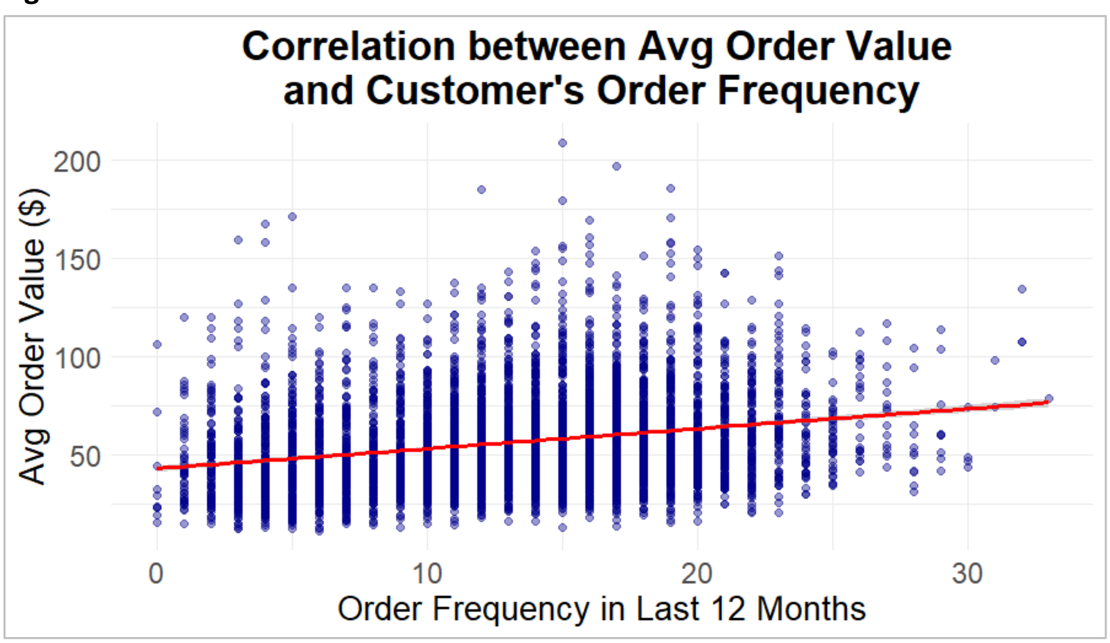


The line and plot diagram tracks the mean delivery (in days) across the different geographical regions.

A peak in Christchurch indicates slightly longer delivery times in this region (0.1 day, or 4% longer) than in the rest. This could be due to a geographical bottleneck, with Christchurch in the South Island, leading to longer transit times or local warehouse inefficiencies.

4. Average Order Value vs Order Frequency

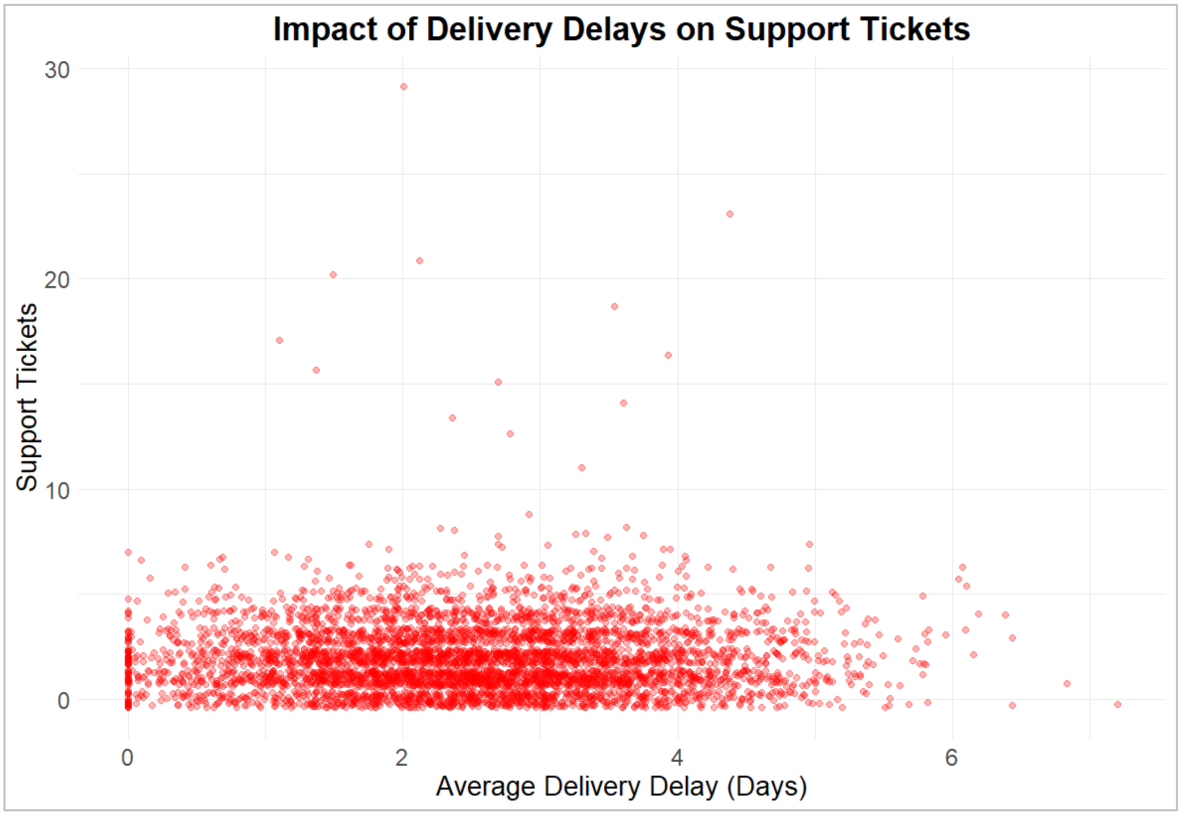
Figure 4



This scatter plot with a regression line examines the correlation between Order Frequency and Average Basket Size.
The positive slope of the regression line shows that high-value customers tend to be high-frequency buyers. Moreover, the spread towards the top highlights a lucrative customer segment.

5. Delivery Delays vs Support Tickets

Figure 5



The jittered scatter plot shows the relationship between the average delivery delay and the number of support tickets requested.
Support tickets remain low until delays exceed 2 days, after which query density increases. Customers with 0-day delays also raise multiple tickets (the vertical spike), suggesting critical issues beyond delivery delays. Interestingly, customers with zero tickets raised despite 4-5 days of delay are seen, suggesting high brand patience.

3. Data Preparation

Variable Selection and Justification

For the clustering model, a comprehensive set of 13 variables was selected, which can be grouped into three distinct dimensions:

- **Engagement and value:** *tenure_months*, *recency_days*, *orders_12m*, and *avg_basket*.
- **Customer preferences:** Channel shares and product category shares.
- **Purchase integrity:** *discount_share* and *return_rate*.

Handling Missing Values

na.omit() was used on the selected 13 variables. Since these behavioural fields were complete, no data was lost, ensuring a clean distance matrix.

Scaling and Transforming Data

All variables were standardized using Z-score normalization, using scale(), to a mean of 0 and an SD of 1.

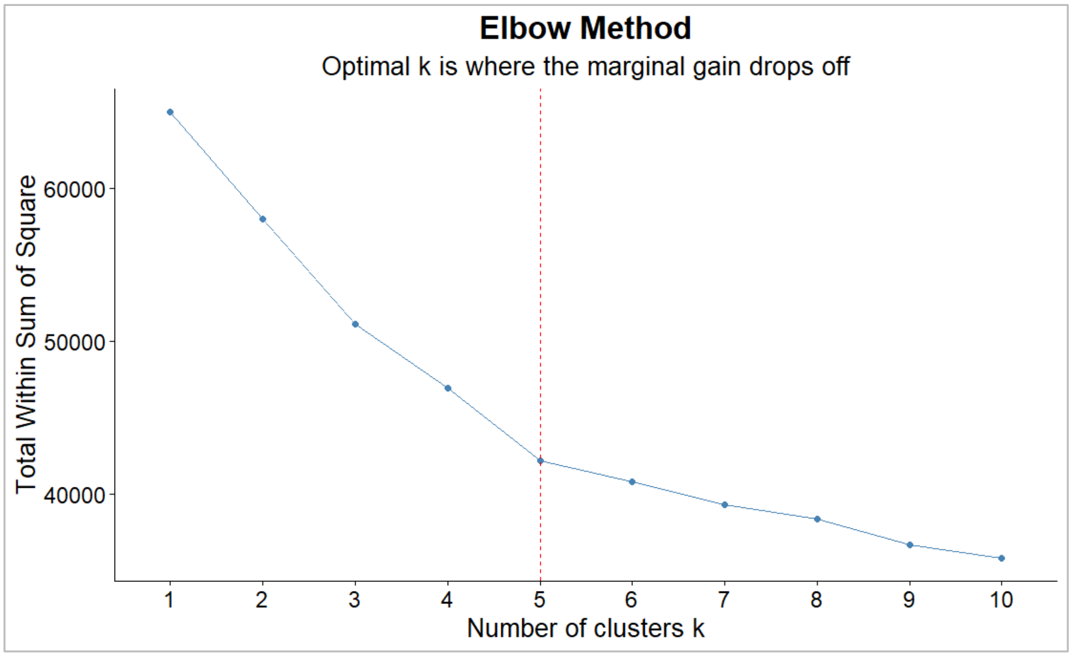
Treatment of Categorical Variables

region, *membership_tier*, and *acquisition_channel* were excluded from the model to prevent mathematical errors in Euclidean distance calculations, but were kept as labels for profiling.

4. Determining the Number of Clusters

Elbow Method

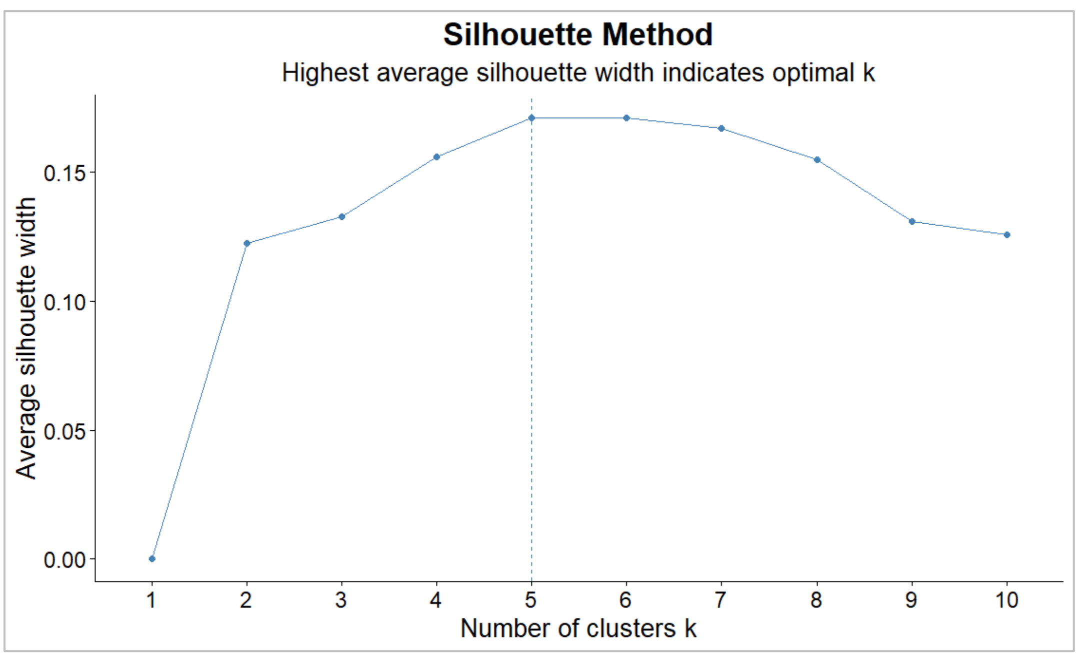
Figure 6



The Elbow plot above shows a sharp drop from k=1 to k=5, beyond which the marginal gain in variance explained diminishes significantly, indicating an elbow at k=5.

Silhouette Method

Figure 7



The average silhouette width peaks at k=5, indicating that this configuration provides the best separation and internal cohesion for this 13-dimensional dataset.

Number of Clusters for the Analysis

Based on the results of both the diagnostic methods, **k=5** was selected as the optimal number of clusters for the analysis. And the selection is justified in terms of statistics, as it provides mathematical backing, and a business perspective, as 5 clusters are neither too many nor too few.

5. Customer Segmentation using K-Means

K-Means with 5 centres was applied to the standardized dataset. Each customer was assigned a cluster label to enable a granular comparison of their behaviour and traits.

Cluster Descriptions

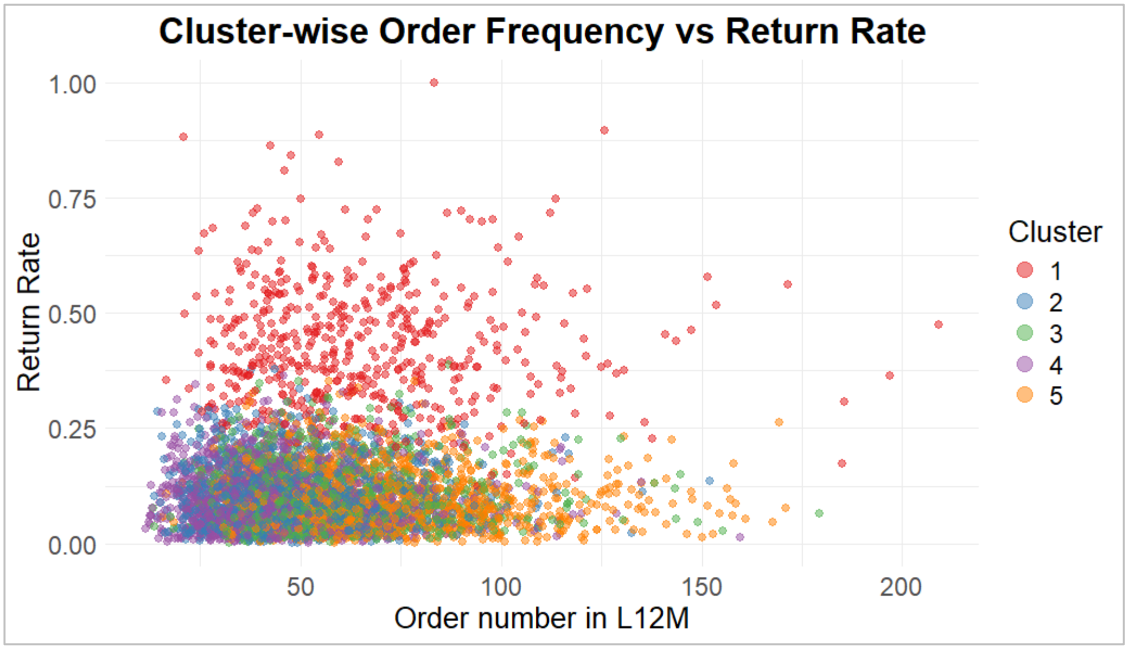
Table 1

Cluster	1	2	3	4	5
avg_basket	65.821	47.578	57.224	43.427	71.63
orders_12m	12.249	13.86	15.511	5.067	17.322
return_rate	0.424	0.101	0.101	0.099	0.097
cat_grocery_share	0.126	0.338	0.157	0.253	0.429
cat_electronics_share	0.342	0.136	0.52	0.243	0.141
cat_fashion_share	0.427	0.392	0.157	0.248	0.105
cat_home_share	0.105	0.134	0.165	0.256	0.324
online_share	0.439	0.439	0.541	0.44	0.403
store_share	0.354	0.356	0.265	0.355	0.415
app_share	0.207	0.205	0.195	0.205	0.182

Cluster 1: High-maintenance Online Fashion Shoppers

High basket size ($65.82) and Fashion preference (42.7%). However, they have an alarmingly high return rate of 42.4%, making them the most expensive segment to service (**Figure 8**).

Figure 8



Cluster 2: Mid-Budget Multi-category Shoppers

Stable customers with moderate baskets ($47.58) and balanced interests in Grocery (33.8%) and Fashion (39.2%). They represent a reliable segment, with a very stable return rate (10.1%).

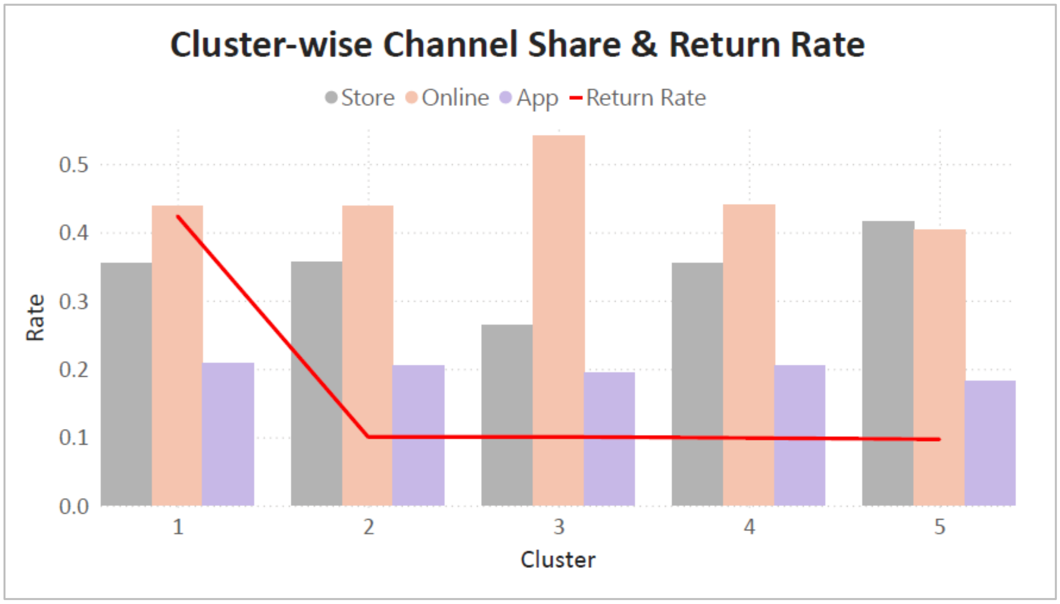
Cluster 3: Online Tech Enthusiasts

The most digitally native group (54.1% Online share) with a 52% focus on Electronics. They shop frequently (15.5 orders/year) with moderate returns (10.1%).

Cluster 4: Low Engagement Generalists

Occasional shoppers with the lowest frequency (5.07 orders/year) and no clear category preference. They are browsers with roughly 25% share across all categories.

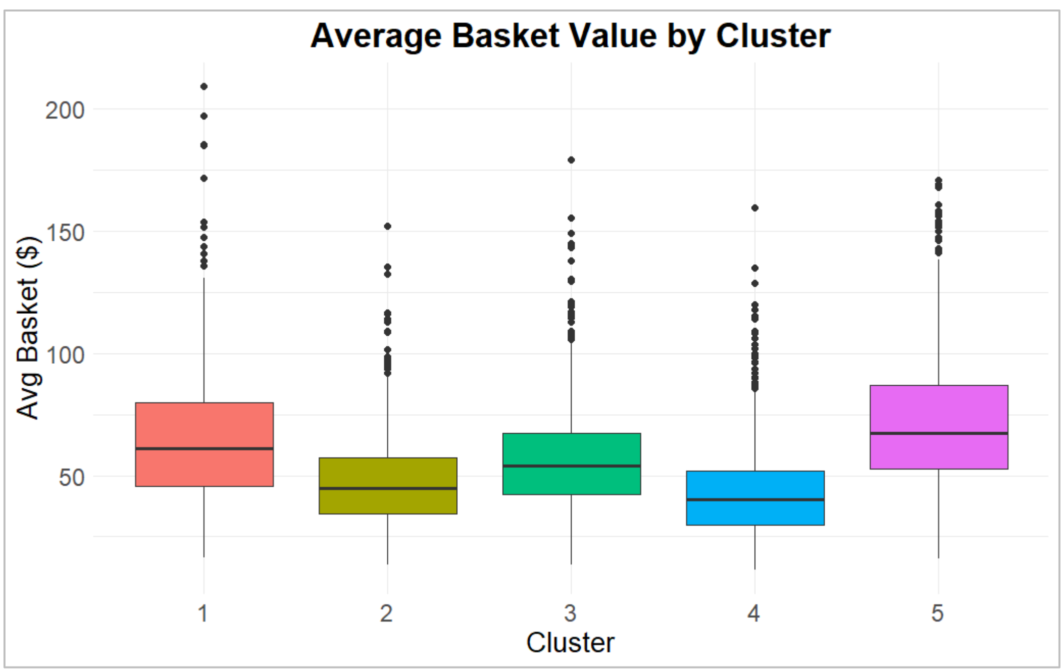
Figure 9



Cluster 5: High-Value Loyal Members

This Star segment has the highest basket ($71.63) (**Figure 10**) and frequency (17.3). They are true omnichannel shoppers (41.5% Store / 40.3% Online) (**Figure 9**) who focus on Grocery and Home goods.

Figure 10



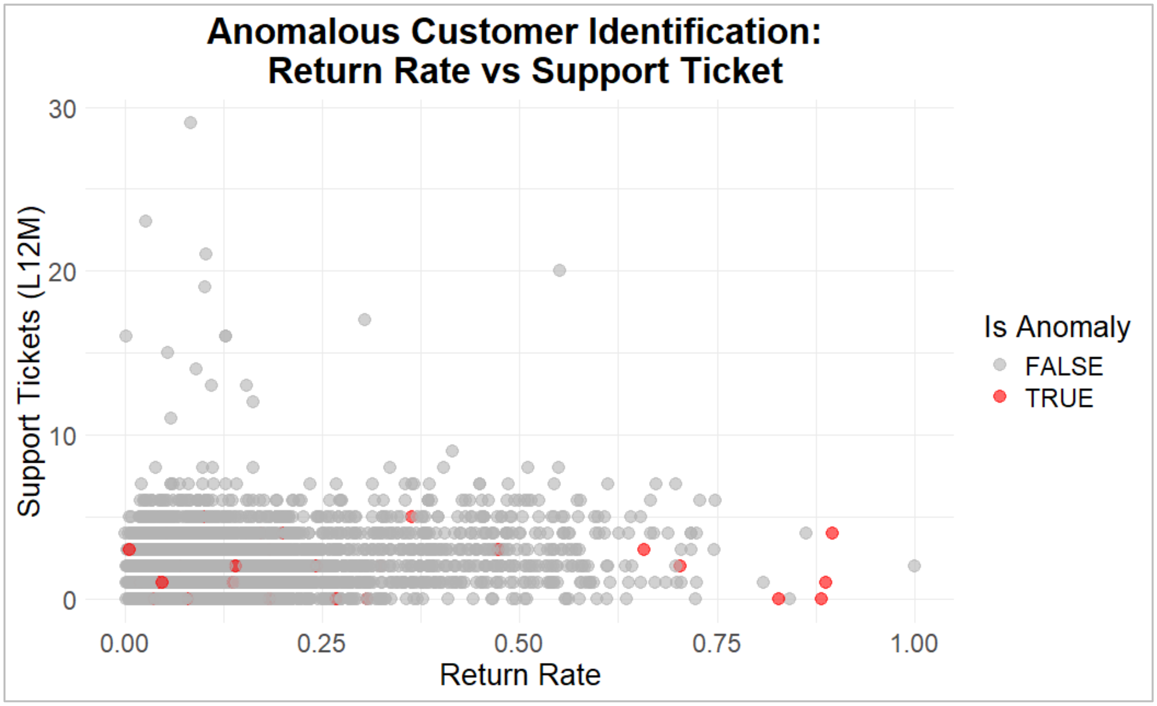
6. Identifying Anomalous Customers

Methodology and Evidence

To identify anomalies, the Euclidean distance for all 5000 observations from their centroids was calculated, and a threshold of twice the mean distance was used.

The scatter plot below shows the outliers. Return rate and Support tickets were chosen because they represent the intersection of operational friction and customer query emergence.

Figure 11



Why are these customers unusual

- The customers on the far right are exceeding 70% return rate. This is unusual, given that the highest returning customers (Cluster 1) have an average of 42%.
- Some anomalies have a significantly high return rate (over 85%) but near-zero support tickets, whereas generally, a high return ratio should be proportional to support tickets.
- Red points near the center indicate that they are not necessarily outliers in return vs tickets. They might be flagged for other extreme behaviour, such as an extreme preference for a single product category or an exceptionally high average basket size.

7. Managerial Insights

1. Mitigating profitability leak in High-Value Fashion (Cluster 1)

Cluster 1 represents a significant revenue source, yet it is the most operationally expensive segment with the highest return rate. They focus heavily on Fashion products and prefer Online shopping. Plus, the presence of outliers with returns exceeding 85% highlights a subset of customers who use the company's logistics as a virtual trial room.

Recommendation: Tech Integration to Facilitate Product Selection

For Cluster 1, the business can shift from investing in volume-based discounts to fit-assurance tools, such as integrating AI-driven size recommendations or AR-based trial features in the App or Online platforms. This could mitigate over 40% revenue loss from returning orders.

2. Driving Growth through Omnichannel Conversion (Clusters 4)

Cluster 4 is the least engaged group, lacking a clear channel or category preference. As seen in Cluster 5, they are the most valuable segment, with their omnichannel behaviour being a key factor. Thus, building multi-channel affinities will yield significant growth from Cluster 4.

Recommendation: Omnichannel Incentives

Loyalty programs that reward channel-crossing can be introduced, such as bonus points or discounts for online-only customers on their first in-store Grocery purchase. Since Cluster 5 shows promising multi-channel behaviour with high Grocery and Home product preferences, Cluster 4 can be targeted towards these channels and categories, leading to order frequency increasing up to 1.5 times while average order value increasing up to 3 times.